

SYNOPSIS

Project Team No: 42

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k.Jahnavikrishna

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Project Title: LeafGuard AI: Predictive Modeling of Plant Stress Using Multisensor Data.

Name of the Guide: Dr.A. SUBRAMANIAN

Abstract

Plants are highly vulnerable to environmental stresses caused by heat, drought, and humidity fluctuations, which significantly reduce growth and yield. This project introduces LeafGuard AI, a predictive framework for plant stress that employs multisensory data and machine learning methodologies. The system uses environmental and physiological factors like temperature, relative humidity (RH), vapor pressure deficit (VPD), volumetric water content (VWC), plant weight, and electrical impedance to check on the health of plants. A representative dataset was utilized, succeeded by preprocessing and feature selection to optimize model efficiency and performance. A combination of Adaboost, decision tree, and random forest, called the Hybrid-AdapTree model, was suggested to make predictions more accurate and easier to understand. The Hybrid-AdapTree model outperformed the others in experimental tests using several regression metrics. It had a R^2 score of 0.999206 for predicting impedance, a low RMSE of 62.34, and a high Pearson correlation of 0.9996. The model was able to predict temperature with a competitive level of accuracy ($R^2 \approx 0.88$). Also, feature selection makes the training process more efficient while keeping the accuracy high and lowering the amount of computation needed. The obtained results prove the reliability and efficiency of the suggested AI-powered framework in accurate processing of sensor data in order to detect plant stress.

Keywords: Plant stress detection ,Temperature ,Humidity ,vapour pressure deficit(VPD),volumetric water content(VWC),AdapTree algorithm, Machine learning.

Specific Contribution

K.Jahnavikrishna:

- Developed and implemented machine learning models such as DT, KNN, MLR, AdaBoost, and MLP.
- Performed data preprocessing, normalization, and dataset splitting.
- Generated heatmaps and feature importance analysis using Mutual Information.

G.Poojitha:

- Developed the Hybrid-AdapTree and Hybrid-AdapTree with Feature Selection models.
- Performed model evaluation using R^2 , RMSE, MAE, and PCC metrics.
- Generated comparison graphs, analyzed results, and calculated Environmental Stress Index (ESI).

Specific Learning

- Learned how to preprocess and analyze sensor-based datasets.
- Learned implementation of machine learning and ensemble models in Python.
- Learned feature selection techniques using Mutual Information.
- Learned model evaluation using regression performance metrics.
- Learned data visualization using heatmaps and comparison plots..

Technical Limitations & Ethical Challenges faced

- Limited dataset size may affect model generalization for different crops and environments.
- Hybrid ensemble models require higher computational time and resources.
- Sensor noise and environmental variations can affect prediction accuracy.
- Real-time deployment requires efficient hardware and IoT integration.
- Ethical use of agricultural data and maintaining data reliability are important challenges.

Name & Signature of the Student**Signature of Guide**

1. G.poojitha

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G.Poojitha
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